

Bioinformatics Applications in Personalized Melanoma Care

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Student name: Afnan Alkhaibari

Student ID: 1718597

1. Introduction

Cutaneous melanoma is considered the most aggressive and lethal form of skin cancer (Xie et al. 2022). This is due to the difficulty of treatment because of the ability of these cancer cells to evolve and resist treatment. Recently, the incidence of melanoma skin cancer has increased, as evidenced by epidemiological data showing a rise in cases among fair-skinned individuals, and clear mortality patterns according to age group and region (Huang et al. 2023, Waseh and Lee 2023). Melanoma is a malignant skin cancer resulting from damage to the pigment-producing cells in the skin, called melanocytes, which are embryonically formed from precursor neural crest cells (Ali, Yousaf, and Larkin 2013). Melanoma is characterised by recurrent driver mutations affecting major oncogenic pathways, particularly BRAF, NRAS, MAPK and PI3K signalling pathways (Smalley 2010). Molecular characteristics also differ between melanoma subtypes, including cutaneous, acral and mucosal melanoma (Miller and Mihm 2006, Schadendorf et al. 2015).

Bioinformatics has become an essential component of modern oncology research and clinical practice, given the biological complexity of melanoma. From prevention to long-term follow-up, it integrates computational analysis of genomic, transcriptomic, imaging, and clinical data to support different stages of patient care. These methods typically rely on high-throughput sequencing technologies, including whole-genome sequencing, whole-exome sequencing, and single-cell RNA sequencing, along with public databases such as TCGA-SKCM, cBioPortal, and GEO (Gao *et al.* 2013).

This essay critically reviews current and emerging applications of bioinformatics across the melanoma care continuum, including prevention, early detection, diagnosis, treatment selection, outcome prediction and surveillance for recurrence. In addition, it discusses the feasibility, limitations and ethical implications associated with these approaches.

2. Current applications of bioinformatics:

2.1. Diagnosis and classification using genomic profiling:

The Cancer Genome Atlas (TCGA) study of cutaneous melanoma represented a key milestone in understanding the molecular landscape of this disease. It demonstrated integrated patterns of driver mutations, copy-number alterations, and gene expression changes that support the classification of melanoma into distinct molecular subtypes. Subsequent whole-genome analyses across major

melanoma subtypes further characterised the landscape of somatic alterations and DNA damage-repair processes, while also distinguishing between UV and non-UV-driven tumours (Hayward *et al.* 2017). These initiatives rely on bioinformatics pipelines that include sequence alignment, variant calling and functional annotation, powered by tools such as GATK, SAMtools, BCFtools and ANNOVAR, and interactive portals such as cBioPortal to navigate multi-omics datasets (Gao *et al.* 2013).

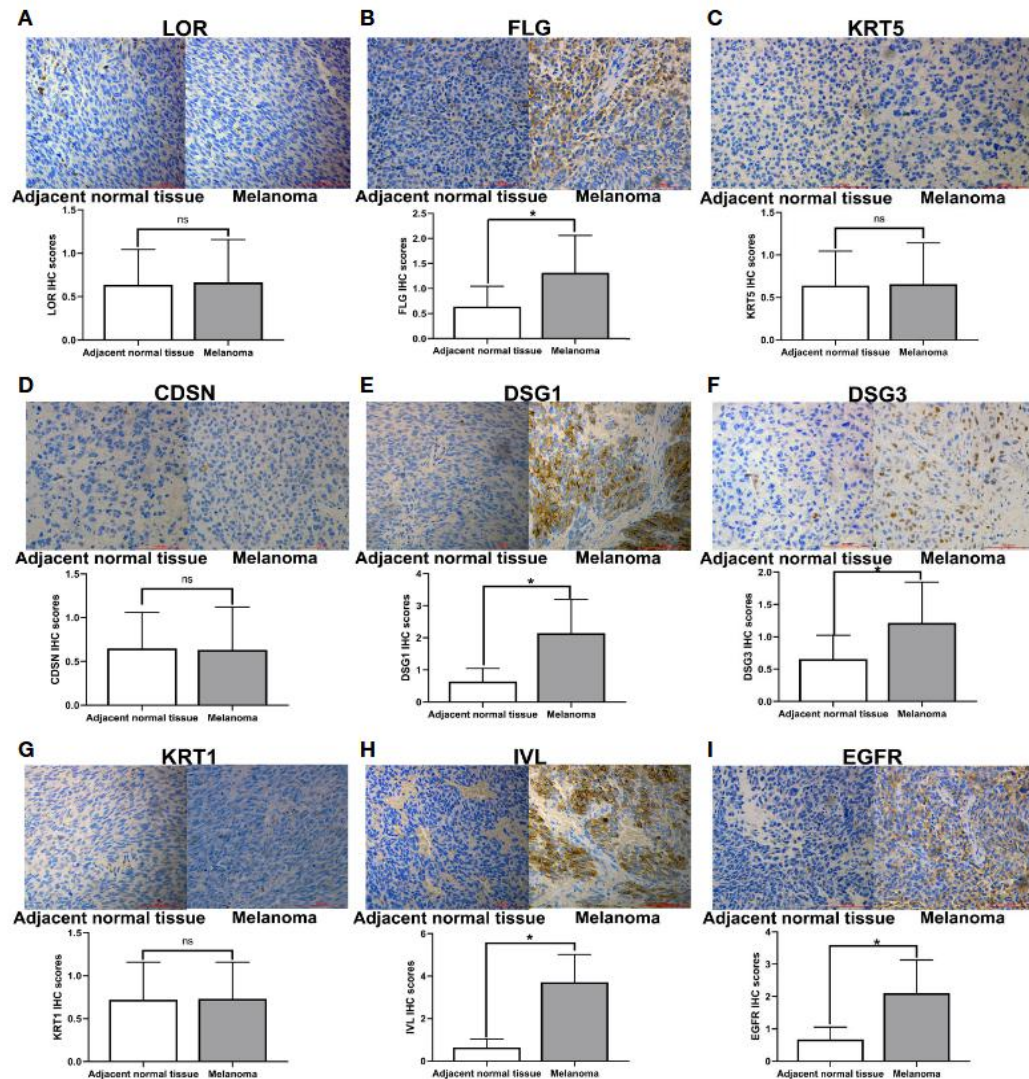
Molecular classification has significant clinical significance. This allows for more precise differentiation of benign and malignant melanocytic lesions, especially in challenging cases, as well as refinement of histopathological subtyping. Furthermore, recognition of specific mutations, such as BRAF V600 and other actionable alterations can directly guide the selection of targeted therapies and combination treatment options. Genomic profiling has thus transitioned from a research tool to an important part of routine melanoma diagnosis and management (Akbari *et al.* 2015, Hayward *et al.* 2017).

2.2. Use of transcriptomic and immune profiling to guide therapy:

The advent of RNA sequencing technologies has provided unprecedented resources for studying the tumour microenvironment and immune cell phenotypes in cutaneous melanoma. For example, a recent single-cell study of metastatic melanoma revealed marked intratumoural heterogeneity and demonstrated that T cells and other immune populations adopt distinct functional states that influence responses to immune checkpoint blockade (Tirosh *et al.* 2016). Bioinformatic analysis of gene expression data has been used to identify genetic patterns linked to either response or resistance to immunotherapies such as anti-PD-1 and anti-CTLA-4, as well as targeted therapies against BRAF and MEK (Zhou *et al.* 2021).

Several immune-related gene signatures have been associated with prognosis and risk of recurrence in melanoma, providing potential biomarkers for risk stratification as well as possible therapeutic targets (Han *et al.* 2020, Zhou *et al.* 2021). These analyses often rely on alignment software such as STAR, along with expression and differential analysis packages like DESeq2, WedgeR, and limma, using path enrichment methods such as GSEA. A comprehensive digital dataset is used for analysis using frameworks like Seurat or Scanpy. (Han *et al.* 2020, Zhou *et al.* 2021). An example of protein-level validation of bioinformatically identified hub genes in melanoma is shown in Figure 1 (Han *et al.* 2020).

Figure 1. Immunohistochemical staining and quantitative analysis of selected hub genes in melanoma tissues and matched adjacent normal tissues (magnification 200×). Adapted from (Han *et al.* 2020) *Frontiers in Oncology*.



2.3. Detection and Diagnosis: Digital Imaging and Dermoscopy:

Deep learning revolutionised how researchers analyse dermoscopic and clinical images in melanoma. In one of the most cited studies, a convolutional neural network trained on millions of images achieved at dermatologist-level classification skin lesions and melanoma (Esteva *et al.* 2017). Later studies directly compared such models with panels of dermatologists using head-to-head dermoscopic classification tasks, finding that deep learning systems can achieve

expert-level performance in many scenarios (Brinker et al. 2019, Huang et al. 2023).

In analogue pathology, platforms like QuPath and HALO are applications that allow analysis of whole-slide images (WSI) to measure mitotic rate, tumour thickness and tumour-infiltrating lymphocytes as well as detect subtle morphological patterns below the limits of routine human detection (Bankhead et al. 2017). Integrated analyses of TCGA data at the cancer-wide level have also shown that cell origin patterns contribute to shaping the molecular classification of multiple tumor types, including melanoma, and can help identify their subtypes (Kather et al. 2020).

2.4. Outcome prediction and surveillance models:

Models such as regularised Cox regression, random forests and other machine-learning approaches have been applied to integrate clinical, histological, genomic and imaging variables into composite predictors of survival and relapse in melanoma (Han *et al.* 2020, Zhou *et al.* 2021). Bioinformatics analysis has also been enhanced using real-world data from electronic health records and cancer registries. Studies have investigated the relationships between clinical features derived from these records and molecular and therapeutic variables to identify factors associated with long-term response to targeted and immunotherapeutic treatments, while also initiating the evaluation of risk-adapted monitoring strategies (Luke et al. 2017). Therefore, the practical implementation of these models depends on machine learning and statistics libraries in R and Python, as well as a robust data warehouse infrastructure, the availability of high-quality clinical data from electronic health records, and standardized outcome definitions (Kelly et al. 2019, Topol 2019).

2.5. Emerging and future applications:

This includes research into polygenic risk scores and personalised prevention. have demonstrated that peripheral blood-based genomic models incorporating common variants can improve melanoma risk prediction over models based solely on phenotypic and environmental variables. Combined with skin type and UV exposure history, such scores may facilitate more personalised prevention and screening (Cust et al. 2018).

These methods depend on bioinformatic pipelines for GWAS analysis and specialised PRS construction/validation tools such as PRSice or LDpred. Nevertheless, the clinical utility and equity of PRS remain to be fully determined, particularly because GWAS datasets suffer from an inherent Eurocentric bias with compelling transferability of risk scores across ancestries (Euesden, Lewis, and O'Reilly 2015).

2.6. Liquid biopsy and circulating biomarkers:

Liquid biopsy is a minimally invasive approach for monitoring melanoma through the analysis of circulating tumour DNA (ctDNA) in blood. Early clinical studies have shown that ctDNA levels can predict response to anti-PD-1 therapy in metastatic melanoma and may detect disease progression before radiological changes become apparent. In addition, treatment-induced changes in ctDNA burden are associated with tumour response and can indicate relapse or treatment failure at an early stage (Wang et al. 2019, Bartolomucci et al. 2025).

Unique molecular marker (UMI)-based data pathways are essential for high-sensitivity circulating tumor DNA (ctDNA) assays, along with variant-detection tools capable of detecting low-allelic mutations, such as VarDict and LoFreq. In this context, bioinformatics plays a pivotal role in distinguishing between true tumor variants and sequence noise, and in linking ctDNA dynamics to clinical and imaging data. Before broader clinical adoption (Sandmann et al. 2017).

Melanoma is increasingly being used as a model system in multimodal AI applications, through the integration of genomic, transcriptomic, dermatological, histological, and clinical data, in what is known as “Melanoma 2.0.” These different layers of data are combined using advanced diagnostic techniques and computational models to improve the assessment of skin lesions and patient classification (Marchetti et al. 2018). Integrated analyses of TCGA data at the cancer-wide level have also shown that cell origin patterns contribute to shaping the molecular classification of multiple tumor types, including melanoma, and can help identify their subtypes (Marchetti et al. 2018).

2.7. Population-level surveillance and near-real-time epidemiology:

methods are being used at a population level to derive near-real-time indicators of melanoma incidence, stage at diagnosis, treatment patterns and survival from national cancer registries, pathology reporting systems and screening programs. Geospatial and temporal models, including data on UV exposure, socioeconomic status and/or healthcare access, can identify high-risk regions and guide targeted prevention campaigns. By linking registry data with engineering molecular profiling and treatment information, it also enables the large-scale evaluation of policy and service interventions in routine practice.

Such applications rely heavily on big-data platforms and geospatial analysis tools paired with Bayesian or other specially grounded statistical frameworks. But they rely on timely and high-quality data capture, and on good systems for sharing and governance across both institutions and jurisdictions (Arnold et al. 2022).

3. Discussion: feasibility, challenges and ethics

3.1. Clinical impact and feasibility:

The effectiveness of several bioinformatics approaches has been verified in research settings; however, their application in the routine care of melanoma remains uneven. While genotyping of targeted mutations such as BRAF has already been included in clinical guidelines (Lee et al. 2017, Luke et al. 2017), Circulating tumor DNA (ctDNA) analysis is still in the early stages of clinical use, particularly in advanced cases of the disease.

Clinical translation of these methods depends on appropriate computational infrastructure and the integration of multidisciplinary teams with electronic health record systems. These components are essential for supporting reliable and scalable AI-assisted clinical decision-making in oncology (Kelly et al. 2019, Topol 2019).

3.2. Technical and data challenges:

One of the most significant obstacles to data quality and homogeneity is technical challenges. This results in limitations and generalizations due to variations in sequencing platforms, sample processing, and bioinformatics pathways. The terminal or mucosa type, in particular, is often underrepresented in public datasets, and representations of non-European populations are frequently of lower quality than HLA-1-restricted imaging genomes, leading to models involving mucosa (Kelly et al. 2019).

Deep learning models for dermatoscopy and digital pathology are typically trained on very large datasets. If these tools are to be trusted and used in the clinic, ongoing external validation efforts will need to be complemented by transparent reporting of training data and assessment metrics for the test suite (Esteva *et al.* 2017, Haenssle *et al.* 2018, Brinker *et al.* 2019).

3.3. Ethical, legal and social considerations:

A wider use of genomic, imaging and clinical data raises significant ethical and legal issues. However, the risk of re-identification of patients, especially those with rare subtypes of tumors or those with unique combinations of clinical and genetic characteristics, remains even after the formal removal of identifying data from databases (Erlich and Narayanan 2014).

Melanoma AI is suffering big time from algorithmic bias and equity issues. One of the challenges encountered in using artificial intelligence for skin cancer diagnosis is algorithmic bias. This is because most models are trained on images of light-skinned populations, which can be less accurate and lead to more diagnostic errors for patients with darker skin, causing delays in their access to appropriate healthcare (Adamson and Smith 2018). Diverse training data and regular review of model performance across different subgroups can contribute to greater fairness

in AI applications. However, challenges remain regarding interpretability and accountability. Physicians and patients need to understand how the models arrive at their decisions, and it's crucial to know who is responsible if an AI-powered decision causes harm. Algorithms like SHAP and LIME can help explain some aspects of how the models work, but they don't eliminate the need for human oversight and ethical considerations (Topol 2019).

4. Conclusion:

In all its various stages of skin cancer care, bioinformatics is of growing importance. For example, from genetic analysis and the search for biomarkers for treatment response to applications that identify biomarkers to artificial intelligence-assisted identification and monitoring systems on the skin cancer scene, AI-aided detection; applications for early warning and monitoring of tumors that monitor genetic status, as well as for free tumor DNA. Nevertheless, the widespread application of these systems in the clinic is still not quite the same for everyone. It should depend on the availability of strong empirical evidence that is supported by prospective studies already made, and the presence of major problems for future research for it to overcome. Finally, integration of bioinformatics into the skin cancer treatment pathway is an ongoing project that requires collaboration between bioinformatics professionals, clinicians, patients, and themselves all with the aim being to promote patient success.

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